Better productivity at entry. A child's adult life changes for two reasons, which separate exactly. Hold the policy fixed at Treatment First and change only who enters adult life: fewer parents had a SUD, so fewer children carry the earnings penalty. This part, ΔE, splits as childhood does: the parent counts the share ωc of it through the charge $D(y, d)$ in (24), so the share $1 - \omega c$ is the earnings externality.

The newborn's own adult life under the policy. Hold who enters fixed at Housing First's entrants, with the earnings penalties they carry, and change only the policy: the same child, on the same starting productivity, drawn under Housing First, lives its adult life under Treatment First, with a voucher easier to draw when sober and a cheaper drug, but with the screen in place. This is the effect of the policy on any adult, including its general-equilibrium effects when πV or the drug price clear. It is exactly the gain the newborn's parent has, because the parent is an entrant too: ΔP, the change in the parent's own lifetime value when the policy switches, holding the distribution of entrants at Housing First's. The child receives it JK years into its life, so it enters the newborn's gain as $\beta JK \Delta P$. This part is not an externality: it would be the same for a child whose parent never used.

What its parent does count. The share ωc of ΔC and of ΔE: the part of the child's gain that its parent weighs in its own choices, so it is not an externality.

ΔP is the change in the parent's value at $j = 1$ of (24) when the policy switches from Housing First to Treatment First, averaged over the households that Housing First's newborns are born into, so that the same households are valued under both policies. ΔA is built from the same value at $j = 1$, the entrant's, discounted by βJK for a newborn, so the split of adult life is exact. For a newborn, $\beta JK \Delta P$ is the change in that value at a given starting productivity drawn under Housing First, so $\Delta E = \Delta A - \beta JK \Delta P$ is what the newborns' better starting productivity under Treatment First adds. In symbols,

$$\Delta W = (1 - \omega c)\, \Delta C \quad \text{childhood externality}$$

$$+ (1 - \omega c)\, \Delta E \quad \text{earnings externality}$$

$$+ \beta JK \Delta P \quad \text{own adult life}$$

$$+ \omega c\, (\Delta C + \Delta E) \quad \text{counted by the parent}$$

. (32)

Table 6 reads each part off the benchmark's welfare curve, one after the other, so that the four sum to the newborn's CEV. Its last row adds the parent's own gain ΔP as a percent of the parent's own lifetime consumption, read off the curve for adult life, undiscounted, since the parent is at $j = 1$ when its child is born. The gain from the newborn's own adult life is smaller in absolute value than its parent's gain, for two reasons: the newborn receives it JK years after birth, discounted, and its lifetime consumption, the base of the percent, includes its childhood as well as its adult life.

Table 6: Where the newborn's gain comes from

Both held πV clears Both clear

The newborn’s gain ΔW, percent of lifetime consumption 2.82 6.45 2.57 Childhood its parent does not count, $(1-\omega c)\Delta C$ 2.20 3.47 1.50 Better productivity at entry its parent does not count, $(1-\omega c)\Delta E$ 1.06 1.00 0.46 The newborn’s own adult life under the policy, $\beta JK\Delta P$ −0.75 1.54 0.43 What its parent does count, $\omega c(\Delta C+\Delta E)$ 0.31 0.44 0.19

The externality, the first two parts, percent of the gain 116 69 76 The parent’s own gain ΔP, percent of the parent’s lifetime consumption −1.19 2.42 0.69

Notes: Treatment First against Housing First in the long run; the columns are the seventh, eighth and last of Table 5. The first row is the CEV of (30). The four rows under it sum to it: each is read off the benchmark’s welfare curve after the ones above it. Childhood is the change in the child’s flow at its weight $\xi-1$, and better productivity at entry the change in its adult value from who enters; each is split into the share ωc its parent counts and the share $1-\omega c$ it does not. Better productivity at entry holds the policy and changes only who enters adult life; the newborn’s own adult life holds who enters and changes only the policy, and equals the parent’s own gain discounted to the child’s birth. The parent’s own gain is the change in its value at $j=1$ when the policy switches from Housing First to Treatment First, with the distribution of entrants held at Housing First’s, for the households Housing First’s newborns are born into, as a percent of the parent’s own lifetime consumption, undiscounted. It is larger in absolute value than the newborn’s own adult life because the newborn receives the same gain JK years later, discounted, and because the newborn’s lifetime consumption, the base of its percent, includes its childhood as well as its adult life.

In the main experiment, the last column of Tables 5 and 6, the newborn gains 2.57 percent of lifetime consumption. Of that, 1.50 percent is childhood and 0.46 percent better productivity at entry, both of which its parent does not count. In all, three quarters of it is an externality. The rest is the newborn’s own adult life under the policy, 0.43 percent, and the 0.19 percent that its parent does count, of childhood and of productivity at entry. Of the externality, the childhood part is the larger in every column: what the child gains at home accrues year by year for JK years, while the earnings penalty reaches it once, at entry. The parents gain as well (the bottom row of the last column), by 0.69 percent of their own lifetime consumption, because the higher πV and the cheaper drug outweigh the option the screen removes: in general equilibrium both the child and the parent gain.

With the drug price and πV held, the first column, the externality is more than all of it, because the newborn’s own adult life is a loss there, of 0.75 percent of lifetime consumption: the screen takes an option from every adult and gives nothing back, and the parents’ own lifetime consumption falls by 1.19 percent. Recycling the savings, the second column, turns the newborn’s own adult life into a gain of 1.54 percent and the parents’ loss into a gain of 2.42 percent, since the vouchers go to households that do not use or are not caught; the cheaper drug then takes part of it back, in the last column.

6.4 The transition

To see what the reform does on the way to the long run, I compute the transition from Housing First to Treatment First. The transition starts from the benchmark, with the reform landing as an unexpected permanent shock at the start of the first year, and I simulate it for seventy years, all the way to the Treatment First steady state. Figure 2 shows what the transition looks like along eight key dimensions.

Panels (a) and (d) show drug use and the share of parents who use, respectively. Drug use falls on impact: 81 percent of its long-run fall happens in the first year. In the first year, the share of parents using drugs falls from 16.35 to 13.57 percent, above its long-run level of 13.24 percent. It stays above its long-run level until year 12, because it counts the parents who have a SUD when the reform lands, whose number falls only slowly (panel (b)). From then on it moves around its long-run level. The screen affects the decisions to go to treatment and to use drugs, both directly and through expectations. A voucher holder that uses risks losing its voucher within a year, and a household that uses risks having a voucher offer revoked later, when it may need one but have a SUD, since a disorder takes years to clear.

Panel (b) shows the share with a SUD. This share falls gradually, because recovery is slow: half of its long-run fall is done by year 11, and nine tenths by year 27. The households that still have a SUD keep demand for the drug up, and the drug price stays above its long-run level for 33 years (panel (c)). The elevated price deters households without a SUD, at every age: in the first decade the number of drug users without a SUD falls by up to 23 percent, against 18 percent in the long run.

Panels (e) and (f) show homelessness and the share of children in a shelter, respectively. Neither rises at any point. Homelessness falls from 0.62 percent of households to 0.60 percent by year 10 and 0.56 percent by year 69, against 0.55 percent in the long run, and the share of children in a shelter falls from 1.03 to 1.00 percent by year 10 and 0.93 percent by year 69, against 0.92 percent in the long run. Both could have risen, since the households that have a SUD when the reform lands face losing their vouchers unless they enter treatment and stay sober. They do not, for three reasons. First, few households lose their vouchers: one with a SUD keeps its voucher by entering treatment and not using, and the screen catches a user with a probability of only 20 percent a year, so at most 0.13 percent of households are excluded from the programme in any year. Second, drug use falls from the first year (panel (a)), so households spend less on drugs and have more left over to house themselves, and fewer acquire a SUD and experience longer unemployment. Third, after the first three years, when it is as low as 6.55 percent, πV stays above its Housing First level of 6.75 percent, so the vouchers the screen frees go to other low-income households. Panel (g) shows the CEV of each cohort by its year of birth. Children born at the reform gain 1.45 percent of lifetime consumption, less than the 2.57 percent of those born in the long run, for two reasons. First, they grow up while the share of households with a SUD is still close to its Housing First level, since it falls only as fast as recovery allows: from 3.83 to 3.47 percent in the first ten years, against 3.08 percent in the long run. More of them therefore spend their childhood with a parent who has the disorder, and more enter adult life with the earnings penalty. Second, πV is lower in the first years than in the long run: between 6.55 and 6.90 percent in the first decade, against 7.80 percent. Part of what Treatment First adds to the budget is the taxes paid by adults who grew up without the earnings penalty, while those adults start working only JK years after the reform. Until then, their parents are less likely to draw a voucher. The gain rises towards its long-run level for children born later, whose parents were themselves born after the reform.

Of the adults alive at the reform, only 11 percent are worse off under it, and even of those who have a SUD when it lands, 33 percent benefit, because the lower drug price more than offsets the loss from the sobriety requirement for some of them. Panel (h) repeats the

comparison in every period of the path, with households in each state weighted as under Housing First. The share worse off stays between 2 and 11 percent, and the share of those with a SUD who are worse off falls from 67 to 50 percent as the price comes down further. Of the households entering adult life at 26, 33 percent are better off in the first year, when πV is still close to its Housing First level, and 99.9 percent by year 69.

6.5 Alternative transitions